

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene Networks

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Materials design often treats strain and chemical substitution as independent control parameters. For quasi-hexagonal (qHP) C₆₀ graphullerene, we show that assumption is quantitatively wrong: (i) strain sensitivity $\alpha = dE/d\epsilon$ changes sign and magnitude with dopant type ($\alpha_{\text{pristine}} \approx -1$, $\alpha_N = -303$, $\alpha_B = +57$ meV/%; Exp. 5); (ii) the synergy order parameter $\mathcal{S}$ [Eq. (1)] reaches $|\mathcal{S}| \sim 7\text{--}24$ meV/atom at tetramer scale under +3% strain—far above the additive limit $\mathcal{S} = 0$; (iii) $|\mathcal{S}| > 0$ persists for $n = 1\text{--}8$ supercells (Exp. 10, 40/40 DFT points). These results define a coupled (ϵ, δ) design space for tuning stability of doped qHP networks rather than separate strain or doping scans.

I. INTRODUCTION

Graphullerenes—two-dimensional networks of qHP C₆₀ molecules—are tunable via substitution and mechanical load [1–4]. Prior work maps doping *or* strain separately; the *joint* response of total energy to (ϵ, δ) under identical protocols is unreported for controlled B/N/P substitution.

Gap. If $E(\epsilon, \delta)$ were additive, a two-scan workflow (strain on pristine + substitution at $\epsilon = 0$) would predict combined states. No metric in the literature tests that residual for qHP C₆₀ across dopants and sizes.

This work delivers three verified outcomes from PBE+DFT-D3 (CP2K): (1) opposite-sign α for N vs. B under the same biaxial grid; (2) non-zero $\mathcal{S}$ at +3% strain quantifying the additive-model error; (3) finite-size scaling to $8 \times \text{C}_{60}$ showing the cross term does not vanish at the largest cells computed. Section II defines $\mathcal{S}$; Sec. III reports numbers; Sec. IV states design implications and limits.

II. COMPUTATIONAL METHODS

Electronic structure calculations use CP2K [5] with PBE and Grimme DFT-D3 (Becke–Johnson damping), matching `experiments/*/inputs/*.inp`. Plane-wave cutoff is 300 Ry (280 Ry for $n \geq 6$ in Exp. 10); SCF convergence uses `EPS_SCF` = 10^{-6} with OT minimization. Basis: DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

Systems. Exp. 5: C₆₀ tetramers (240 atoms), six-fold B/N/P substitution, biaxial strain $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ (rigid cell scaling, fixed fractional coordinates). Exp. 7: same tetramers, PDOS and gap analysis at $\epsilon \in \{-5, 0, +5\}\%$. Exp. 10: $n \times \text{C}_{60}$ ($n = 1\text{--}8$) at $\epsilon = 0$ and +3% (40 converged calculations).

TABLE I. Exp. 5 tetramer results (PBE+D3). [verified: `table1.verification.json`]

System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	−1363.551	−1363.551	−1.0	—
B-doped	−1328.490	−1328.409	+56.9	+159.0
N-doped	−1414.381	−1414.405	−303.2	−230.5
P-doped	−1370.496	−1370.489	−19.1	−31.5

Observables. α from linear fits to Exp. 5 energies [Table I]. $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is a relative substitution energy (no μ reference).

We quantify non-additive coupling with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ means the strained, doped state is *more* stable than the two-scan sum predicts (cooperative stabilization), whereas $\mathcal{S} > 0$ means it is *less* stable (cooperative destabilization). Exp. 10 values are from `c/simukit-sdc (sdc_exp10_results.json)`.

III. RESULTS

A. Dopant-dependent strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%), B (+56.9 meV/%), P (−19.1 meV/%). Figure 1 links tetramer *geometry* to mechanics: (a–b) B/N/P structure morphologies and in-plane cell expansion; (c–d) facet regression of opposite α for N vs. B.

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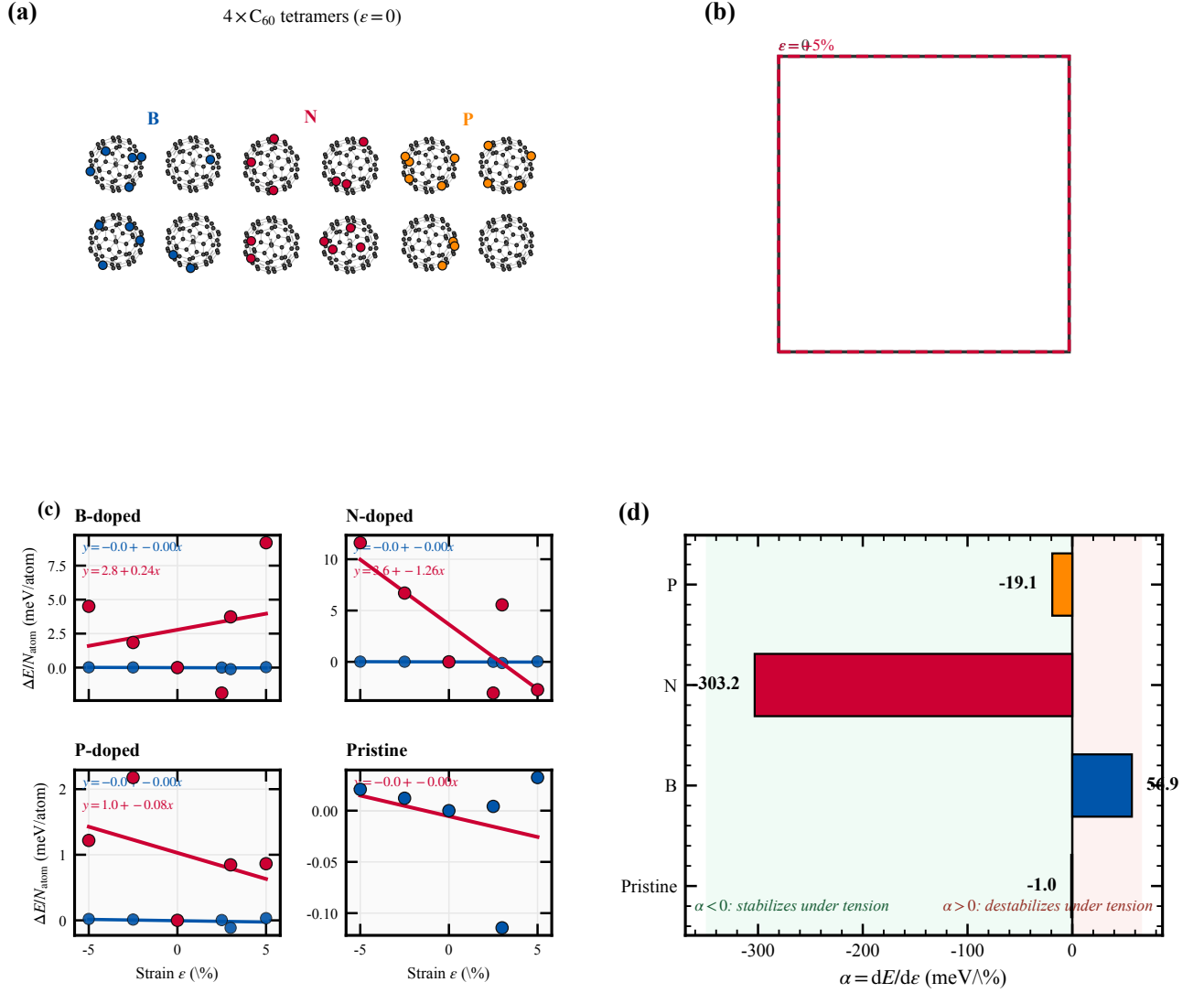


FIG. 1. Structure-mechanics coupling (Exp. 5). (a) B/N/P tetramer top views ($\epsilon = 0$; dopant sites highlighted). (b) Biaxial cell expansion ($\epsilon = 0$ vs. $+5\%$). (c) Facet regression of $\Delta E/N$ vs. ϵ (blue: pristine; red: dopant). (d) α by dopant.

B. Electronic structure and localization

Figure 2 connects mechanics to π electronics (Exp. 7) and carrier localization (Exp. 4). Panel (a) is a gap-regime phase map in (ϵ, δ) with critical near-gap B at $\epsilon = 0$; (b) stacks π -PDOS curves (XRD-style); (c) band-edge morphologies; (d–e) XPS-style deconvolution at $+5\%$; (f) IPR/ J schematic for pristine vs. B under $+3\%$ strain (two verified Exp. 4 points only).

C. Non-additive synergy: tetramer scale and size scaling

Figure 3 maps the non-additive design space: (a) strain-only shift vs. $\mathcal{S}$; (b) mechanical-non-additive phase map (α vs. $\mathcal{S}$ at $n = 4$, $+3\%$); (c) floating-bar de-

composition (sequential vs. coupled DFT; red = synergy gap); (d) $\mathcal{S}(n)$ scaling.

IV. DISCUSSION

Design implications. (1) *Do not rank dopants at $\epsilon = 0$ alone:* N is most stable at zero strain but its α implies further stabilization in tension, whereas B becomes less stable under the same load. (2) *Use $\mathcal{S}$ to screen (ϵ, δ) :* at $+3\%$ strain and tetramer size, P ($|\mathcal{S}| \approx 24$ meV/atom) shows the strongest non-additive correction; single-knob scans would mis-estimate its energy by orders of magnitude larger than N or B at the same conditions. (3) *Coupled maps beat sequential scans:* Exp. 10 required 40 converged cells but yields a reusable non-additivity metric at each size—cheaper than exhaustive transport or geometry paths for ranking stability trends.

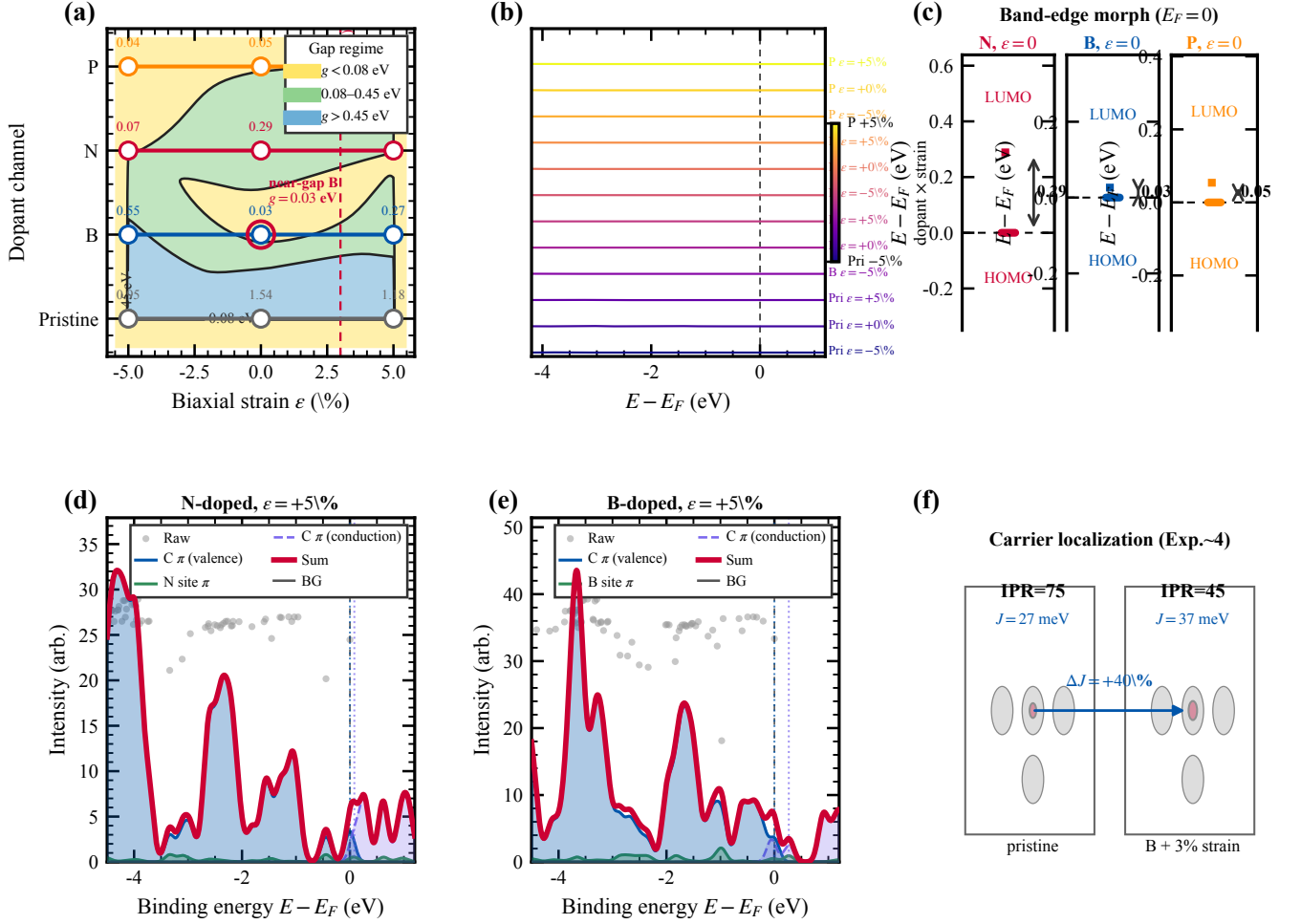


FIG. 2. Electronic morphology (Exp. 7, Exp. 4). (a) Gap-regime phase diagram (strain $\times$ dopant). (b) Stacked π -PDOS waterfall. (c) Band edges at $\epsilon = 0$. (d-e) PDOS deconvolution (N, B). (f) IPR/ J coupling schematic (two-point audit).

Mechanistically, opposite α signs correlate with dopant-dependent gap closing (Fig. 2b,c): B-doped tetramers approach a near-zero gap at $\epsilon = 0$ whereas N remains gap-modulated under tension—consistent with π -electron count differences, though relaxed-ion and hybrid-functional checks are still needed. At the largest supercell ($n = 8$), N switches from cooperative to anti-cooperative non-additivity ($\mathcal{S} = +2.4$ meV/atom at $n = 6$ vs. -2.3 at $n = 8$), indicating that finite-size extrapolation of $\mathcal{S}$ must be dopant-specific rather than universal.

For transport-relevant observables, the two-point Exp. 4 audit shows delocalization (IPR 75 $\rightarrow$ 45) with enhanced nearest-neighbor coupling (J 27 $\rightarrow$ 37 meV, +40%) under B doping at +3% strain—a trend consistent with tighter π overlap but insufficient to claim a polaron-to-band transition without additional IPR/ J grids and Marcus-network mobility calculations.

Limitations. Rigid strain; E_f without chemical potentials; N $\mathcal{S}$ sign change by $n = 8$; PBE+D3 only; Exp. 4 provides only two IPR/ J points—delocalization trend is

illustrated but band-transport predictions remain pending.

V. CONCLUSIONS

qHP C₆₀ graphullerene violates additive strain-doping superposition in total energy. The effect is large (opposite α for N vs. B), quantifiable ($\mathcal{S} \neq 0$ up to ~ 24 meV/atom at tetramer scale), and persistent to $8 \times C_{60}$. For materials design, (ϵ, δ) should be treated as coupled variables; α and $\mathcal{S}$ provide auditable metrics from the DFT workflow archived in this repository.

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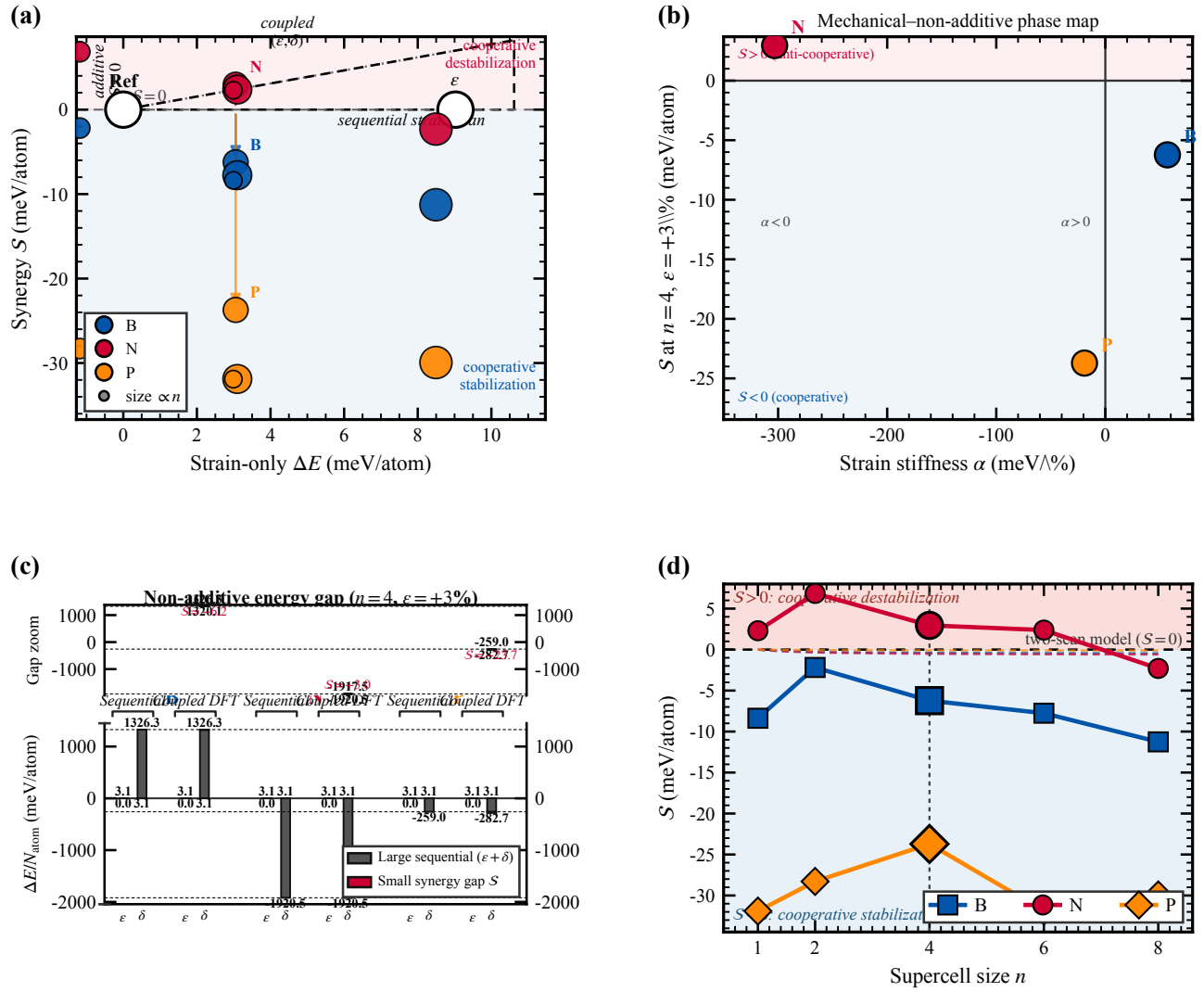


FIG. 3. Non-additive design space (Exp. 5, Exp. 10). (a) Design-space triangle. (b) α - S phase map at tetramer scale. (c) Floating-bar non-additive gap (sequential vs. coupled). (d) $S(n)$ to $n=8$.

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